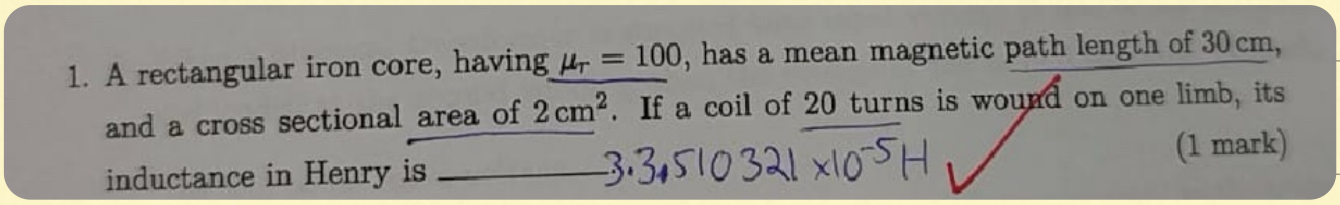
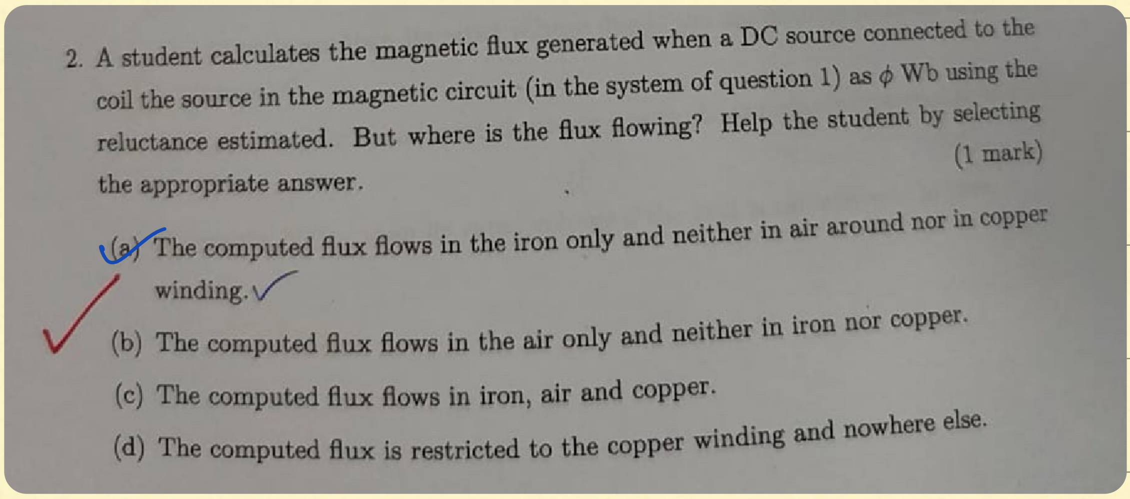


1. A rectangular iron core, having $\mu_r = 100$, has a mean magnetic path length of 30 cm, and a cross sectional area of $2\,\text{cm}^2$. If a coil of 20 turns is wound on one limb, its inductance in Henry is $3.3{,}510321 \times 10^{-5}$ H ✓ (1 mark)

$$L = \frac{N^2 \mu_0 \mu_r A}{l} = \frac{400 \times 100 \times 4\pi \times 10^{-7} \times 2 \times 10^{-4}}{30 \times 10^{-2}} = 3.351032 \times 10^{-5}\,H$$



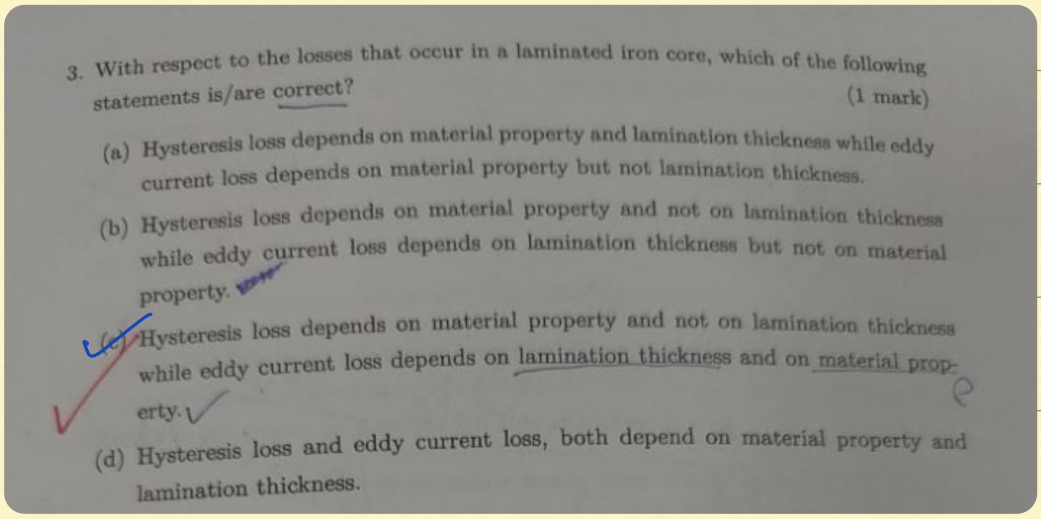
2. A student calculates the magnetic flux generated when a DC source connected to the coil the source in the magnetic circuit (in the system of question 1) as ϕ Wb using the reluctance estimated. But where is the flux flowing? Help the student by selecting the appropriate answer. (1 mark)

✓ (a) The computed flux flows in the iron only and neither in air around nor in copper winding. ✓

(b) The computed flux flows in the air only and neither in iron nor copper.

(c) The computed flux flows in iron, air and copper.

(d) The computed flux is restricted to the copper winding and nowhere else.



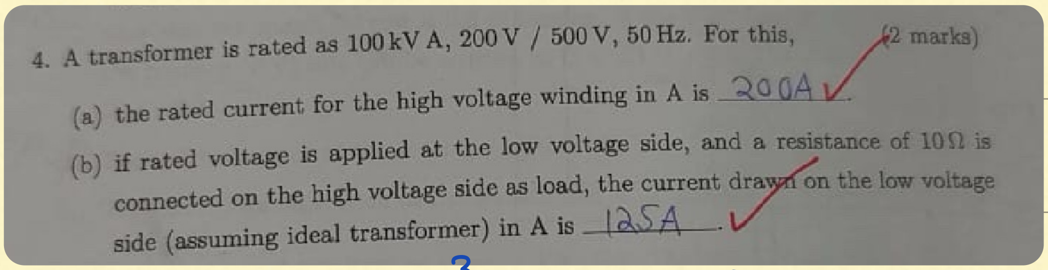
3. With respect to the losses that occur in a laminated iron core, which of the following statements is/are correct? (1 mark)

(a) Hysteresis loss depends on material property and lamination thickness while eddy current loss depends on material property but not lamination thickness.

(b) Hysteresis loss depends on material property and not on lamination thickness while eddy current loss depends on lamination thickness but not on material property.

✓ (c) Hysteresis loss depends on material property and not on lamination thickness while eddy current loss depends on lamination thickness and on material property. ✓

(d) Hysteresis loss and eddy current loss, both depend on material property and lamination thickness.



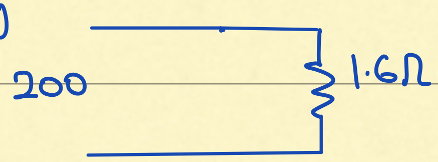
4. A transformer is rated as 100 kV A, 200 V / 500 V, 50 Hz. For this, (2 marks)

(a) the rated current for the high voltage winding in A is 200A ✓.

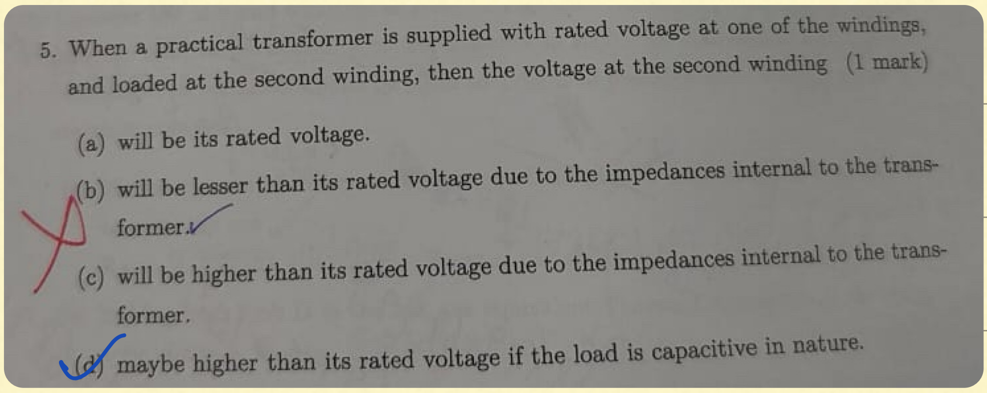
(b) if rated voltage is applied at the low voltage side, and a resistance of 10 Ω is connected on the high voltage side as load, the current drawn on the low voltage side (assuming ideal transformer) in A is 125A. ✓

a) $I_1 = 100 \times 10^3 / 500 = 200\,A$

b)



$1.6\Omega \Rightarrow I_L = 125\,A$



5. When a practical transformer is supplied with rated voltage at one of the windings, and loaded at the second winding, then the voltage at the second winding (1 mark)

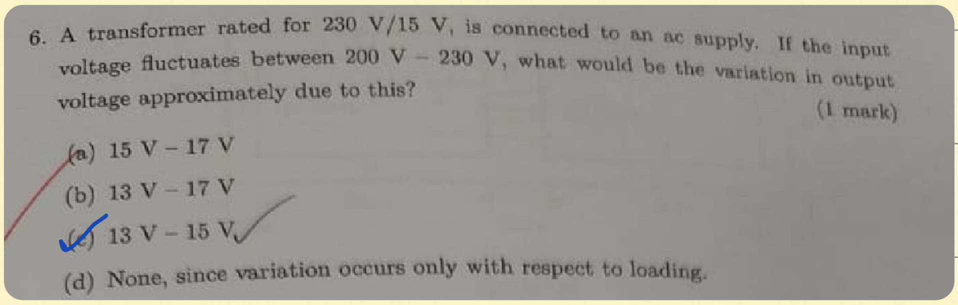
   (a) will be its rated voltage.

   (b) will be lesser than its rated voltage due to the impedances internal to the transformer. ✓

   (c) will be higher than its rated voltage due to the impedances internal to the transformer.

   [✓ (d)] maybe higher than its rated voltage if the load is capacitive in nature.

$V_2 = E_2 - I_2(R+jX)$



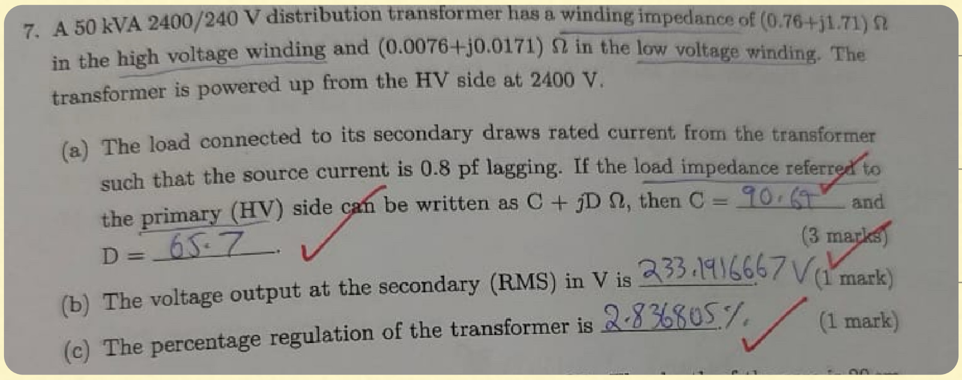
6. A transformer rated for 230 V/15 V, is connected to an ac supply. If the input voltage fluctuates between 200 V – 230 V, what would be the variation in output voltage approximately due to this? (1 mark)

   (a) 15 V – 17 V

   (b) 13 V – 17 V

   [✓ (c)] 13 V – 15 V ✓

   (d) None, since variation occurs only with respect to loading.



7. A 50 kVA 2400/240 V distribution transformer has a winding impedance of (0.76+j1.71) Ω in the high voltage winding and (0.0076+j0.0171) Ω in the low voltage winding. The transformer is powered up from the HV side at 2400 V.

   (a) The load connected to its secondary draws rated current from the transformer such that the source current is 0.8 pf lagging. If the load impedance referred to the primary (HV) side can be written as C + jD Ω, then C = 90.64 and D = 65.7. ✓ (3 marks) ✓

   (b) The voltage output at the secondary (RMS) in V is 233.1916667 V (1 mark) ✓

   (c) The percentage regulation of the transformer is 2.836805 %. ✓ (1 mark)

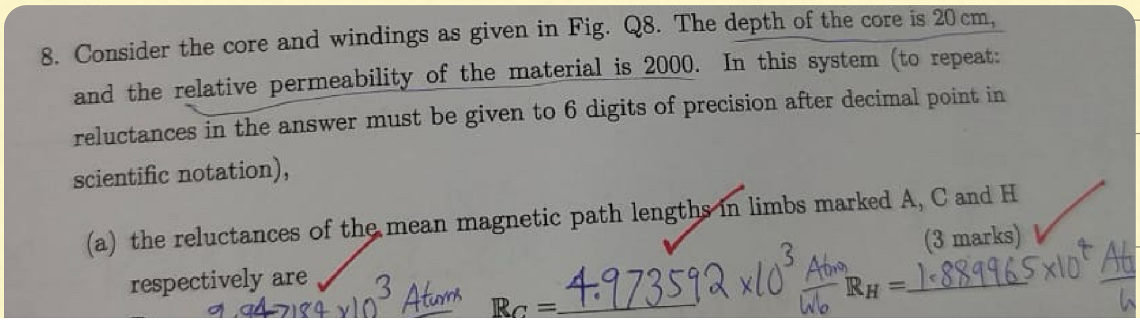
a) $I_1 = \frac{50\times10^3}{2400} = 20.833333 \Rightarrow \bar{I}_1' = 20.833333 \angle -36.87^\circ$

$\Rightarrow Z = \frac{V_1}{I_1'} = 115.2 \angle 36.87^\circ = 92.16 + 69.12j$

$Z_{eq_1} = Z_p + a^2 Z_s = 1.52 + 3.42j \Rightarrow Z_{load} = 90.64 + 65.7j$

b) $\frac{2400 - (20.833\angle -36.87)(1.52+3.42j)}{10} = 233.191667$ V

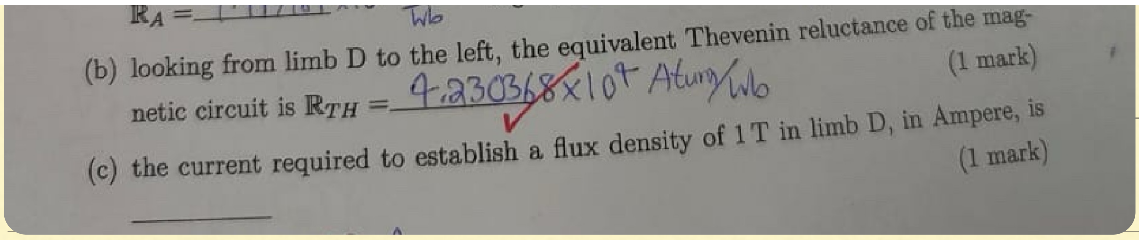
c) $1 - \frac{233.191667}{240} = 2.836805\%$



8. Consider the core and windings as given in Fig. Q8. The depth of the core is 20 cm, and the relative permeability of the material is 2000. In this system (to repeat: reluctances in the answer must be given to 6 digits of precision after decimal point in scientific notation),

   (a) the reluctances of the mean magnetic path lengths in limbs marked A, C and H respectively are ✓ (3 marks) ✓

   9.947184×10^3 Aturns ✓ $\mathbb{R}_C$ = 4.973592×10^3 $\frac{\text{Atrn}}{\text{Wb}}$ $\mathbb{R}_H$ = 1.889965×10^4 At



$\mathbb{R}_A =$ ______ Wb

(b) looking from limb D to the left, the equivalent Thevenin reluctance of the magnetic circuit is $\mathbb{R}_{TH} =$ 4.230368×10⁴ Aturn/Wb (1 mark)

(c) the current required to establish a flux density of 1 T in limb D, in Ampere, is (1 mark)

______

a)

$$R_A = \frac{50\times10^{-2}}{4\pi\times10^{-7}\times2\times10^{3}\times20\times10\times10^{-4}}$$

$$= 9.947183\times10^{3} \text{ A turns/wb}$$

$$R_c = \frac{R_A}{2} = 4.973591\times10^{3} \text{ Aturns/wb}$$

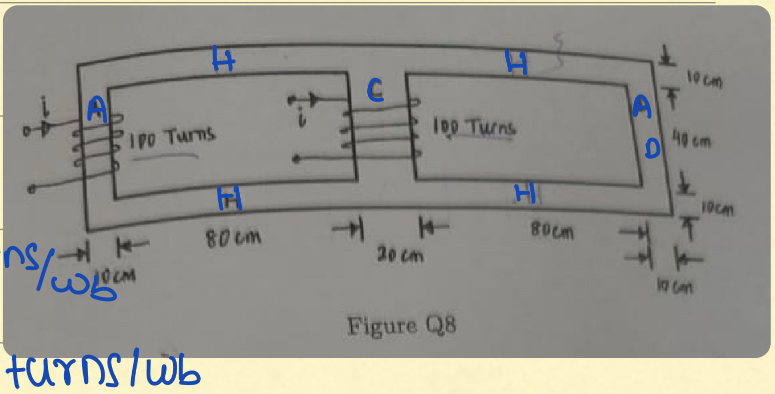


Figure Q8

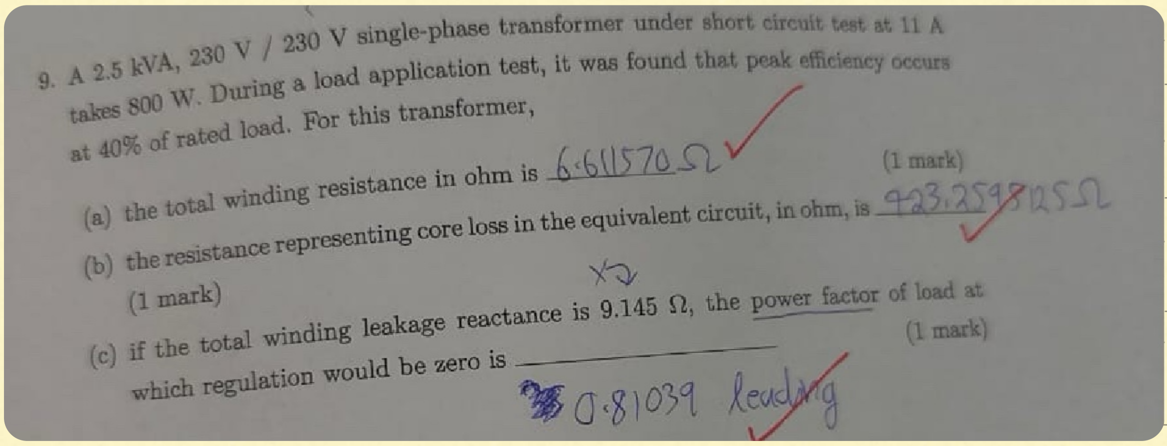
$$R_H = R_A\times\frac{95}{50} = 1.889965\times10^{4} \text{ Aturns/wb}$$

b) $(2R_H + R_A) \parallel R_C + 2R_H = 4.230366\times10^{4}$ Aturns/wb

$\underset{4.774648\times10^{4}}{\downarrow}$ $\quad \underset{0.450438}{\searrow}$

c) $R_{total} = R_{th} + R_D = 4.230366 + 0.994718 = 5.225086\times10^{4}$

$$100\times I = 1\times0.02\times5.225086\times10^{4} \Rightarrow I = 10.450172\text{ A}$$



9. A 2.5 kVA, 230 V / 230 V single-phase transformer under short circuit test at 11 A takes 800 W. During a load application test, it was found that peak efficiency occurs at 40% of rated load. For this transformer,

(a) the total winding resistance in ohm is 6.611570 Ω (1 mark)

(b) the resistance representing core loss in the equivalent circuit, in ohm, is 423.2598125 Ω (1 mark)

(c) if the total winding leakage reactance is 9.145 Ω, XΩ the power factor of load at which regulation would be zero is (1 mark)

0.81039 leading

$$i_{rated} = \frac{2500}{230}$$

$$P_{cu} = (10.86)^2(6.61)$$

$$P_{Fe} = 40\%^{2}\, P_{cu}$$

$$P_c = \frac{V_{rated}}{P_{Fe}}$$

a) $P_{cu} = 800\text{W}$ $\qquad 800 = I_2^2 R_{eq} \Rightarrow R_{eq} = \frac{800}{11^2} = 6.61\,\Omega$

b) $P_{core} = 125\text{ W} \Rightarrow R_c = \frac{230^2}{125} = 423.3\,\Omega$

c) $R\cos\phi = X\sin\phi \Rightarrow \tan\phi = \frac{R}{X} = 0.7228 \Rightarrow \cos\phi = 0.81$ leading